Some Open Source DIY Oxygen Concentrators

If you have been wondering why do-it-yourself (DIY) and open source communities are not proposing any solution for the oxygen concentrators that are in great demand due to the ill effects of Covid-19, here are some of the best open source DIY designs that have been recently introduced by them for setting up or manufacturing oxygen concentrators

Ashwini Kumar Sinha

The outbreak of the second wave of Covid-19 has led to a huge shortage of oxygen concentrators in India. Thankfully, some techies are paving the way for quick manufacture of oxygen concentrators with their open source do-it-yourself (DIY) designs to save lives.

There are some open source Indian designs too, for which parts are easily available, thereby reducing the cost of their basic assembly to as low as US$ 300 (approx. 22,000 rupees)! Here are a few that EFY team has found for you. Those looking for further information about these concentrators are advised to contact the developers directly.

MARUT

Claimed to be India`s first open source oxygen concentrator, Marut has been designed using easily available parts that can be obtained from any hardware shop in India. It also gives the flexibility to choose between a DC-DC converter (to get 12V to 24V, or 12V SMPS) and a 220V solenoid. According to availability in the market, it uses Pressure Swing Adsorption technology for separating nitrogen and oxygen for concentration. This concentrator can deliver oxygen with a concentration of 90 to 92% at a flow rate of ten litres per minute while consuming only 750 watts.

For details: *https://www.technido.com/marut*

ReapRap

ReapRap delivers oxygen with a concentration of 90% (flow rate details not provided). Unlike other oxygen concentrator designs, it uses a MOSFET that drives the solenoid valve. The design is compact; the whole thing is about the size of a shoe-box without the compressor.

For details: *https://reprapltd.com/open-source-oxygen-concentrator*